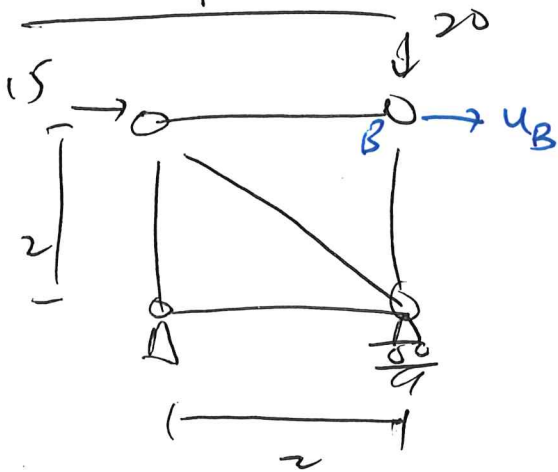


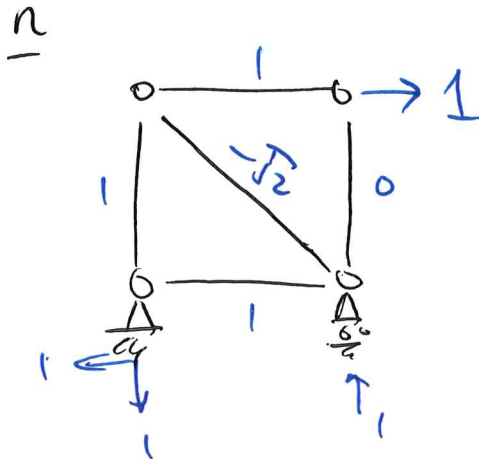
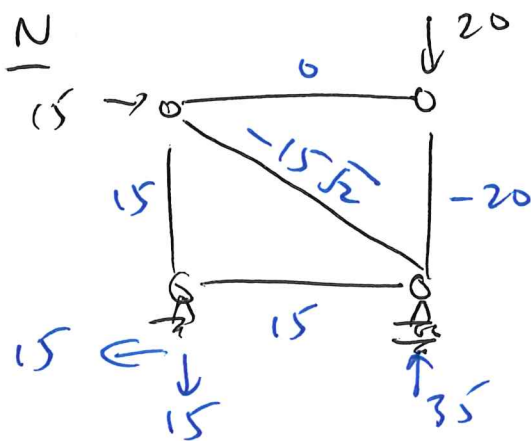
Example 1

wk 6 - Thu 2025
Afternoon

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$$u_B = \sum \frac{nNL}{EA}$$



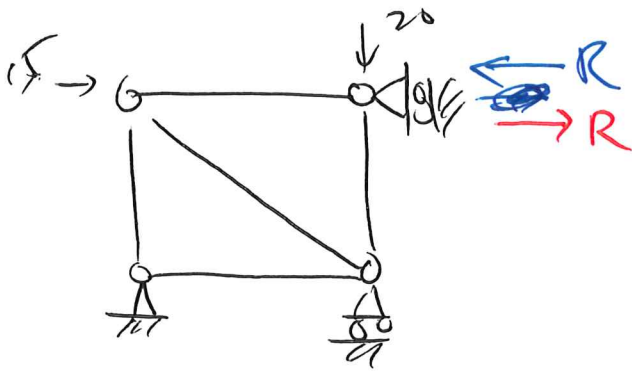
$$u_B = \frac{1}{EA} \sum nNL$$

$$= \frac{1}{EA} \left(1 \times 0 \times 2 + 0 \times (-20) \times 2 + 1 \times 15 \times 2 + 1 \times 15 \times 2 + (-\sqrt{2}) \times (-15\sqrt{2}) \times 2\sqrt{2} \right)$$

$$= \frac{1}{EA} (30 + 30 + 60\sqrt{2}) \approx \frac{145}{EA}$$

Example 2

2/



$$R = \frac{-\sum \frac{nNL}{EA}}{\sum \frac{n^2L}{EA}}$$

Note: From Example 1: $\sum \frac{nNL}{EA} = \frac{145}{EA}$

$$\sum \frac{n^2L}{EA} = \frac{1}{EA} \sum n^2L$$

$$= \frac{1}{EA} (1^2 \times 2 + 0^2 \times 2 + 1^2 \times 2 + 1^2 \times 2 + (-\sqrt{2})^2 \times 2\sqrt{2})$$

$$= \frac{1}{EA} (6 + 4\sqrt{2})$$

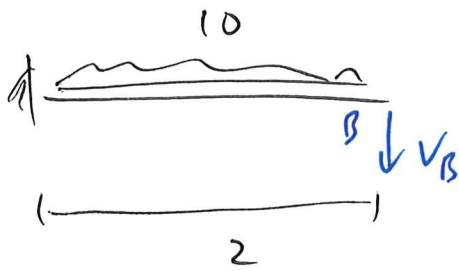
$$R = \frac{-145/EA}{(6+4\sqrt{2})/EA} = -12.43 \quad (\leftarrow)$$

Total axial forces in the members

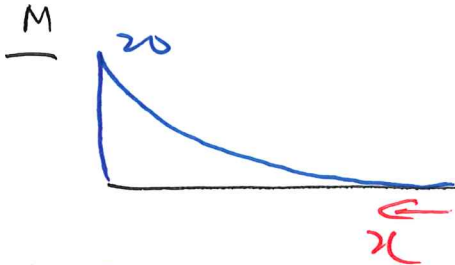
$$N + nR$$

Example 3

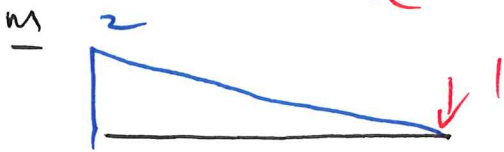
3/



$$V_B = \int_0^L \frac{mM}{EI} dx$$



$$M(x) = -10 \cdot x - \frac{x^2}{2} = -5x^2$$



$$m(x) = -x$$

$$V_B = \frac{1}{EI} \int m M dx$$

$$= \frac{1}{EI} \int_0^2 (-x) \cdot \frac{(-5x^2)}{5x^3} dx = \frac{20}{EI} (\downarrow)$$

$$V_B = \frac{1}{EI} \int_0^2 \left(\text{triangle with height 2, base 2} \right) \cdot \left(\text{triangle with height 20, base 2} \right) dx$$

$$= \frac{1}{EI} \cdot \frac{1}{4} \times 2 \times 20 \times 2$$

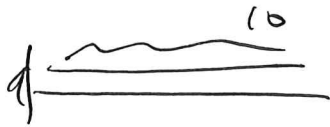
$$= \frac{20}{EI}$$

Example 4

4/

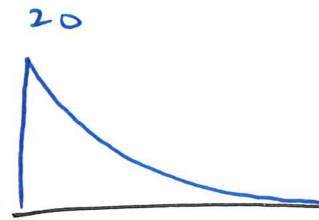


l

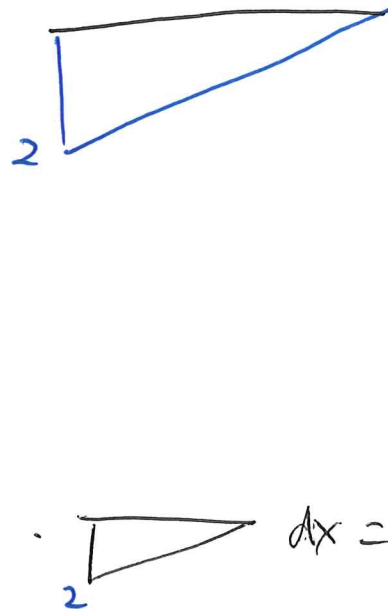


l

M



m



$$R = \frac{- \int_0^l \frac{mM}{EI} dx}{\int_0^l \frac{m^2}{EI} dx}$$

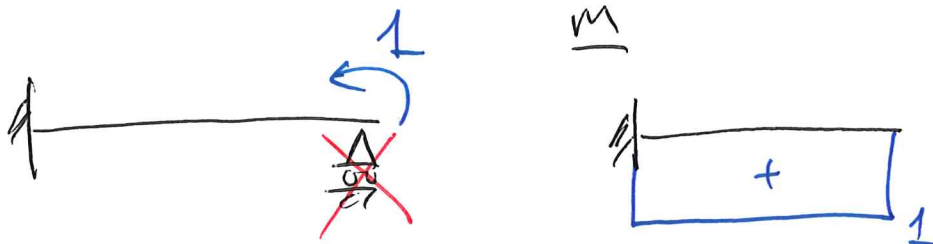
$$\int_0^l \frac{mM}{EI} dx = \frac{1}{EI} \int_0^l \left(\text{triangle with height 20} \right) \cdot \left(\text{triangle with height 2} \right) dx = - \frac{20}{EI}$$

$$\begin{aligned} \int_0^l \frac{m^2}{EI} dx &= \frac{1}{EI} \int_0^l \left(\text{triangle with height 2} \right) \cdot \left(\text{triangle with height 2} \right) dx = \\ &= \frac{1}{EI} \times \frac{1}{3} \times 2 \times 2 \times 2 = \frac{8}{3EI} \end{aligned}$$

$$R = - \frac{-20/EI}{8/3EI} = 7.5 (\uparrow)$$

Total BM : $M + mR$

5/



$$\theta_B = \frac{1}{EI} \int_0^L \left[\text{rectangular diagram} \right] \cdot \left[\text{triangular diagram} \right] dx$$

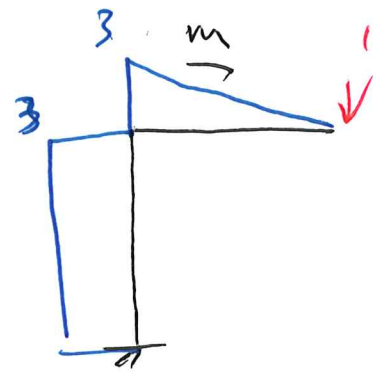
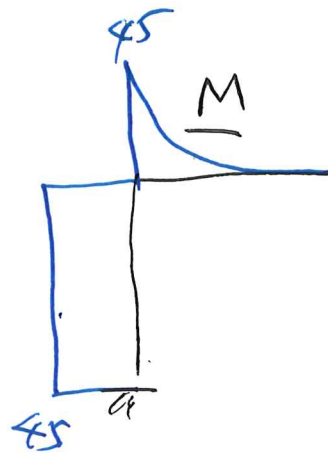
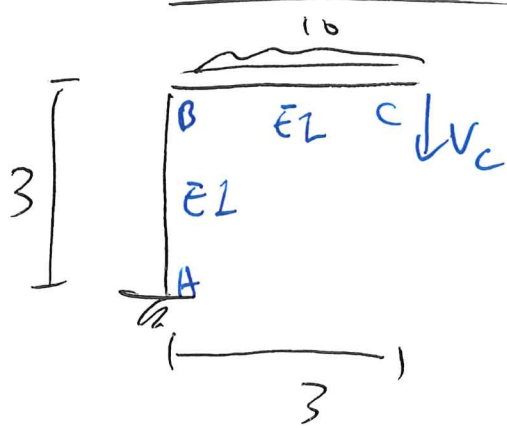
$$= \frac{1}{EI} \left(\int_0^L \left[\text{rectangular diagram} \right] \cdot \left[\text{triangular diagram} \right] dx \right.$$

$$\left. + R \times \int_0^L \left[\text{rectangular diagram} \right] \cdot \left[\text{triangular diagram} \right] dx \right)$$

$$= \frac{1}{EI} \times \left(\frac{1}{3} \times 1 \times (20) \times 2 + R \times \frac{1}{2} \times 1 \times 2 \times 2 \right)$$

$$= \frac{5}{3EI} \quad (\leftarrow)$$

Example 5



6/

$$V_c = \sum \int \frac{mM}{EI} dx$$

$$= \frac{1}{EI} \sum \int m M dx$$

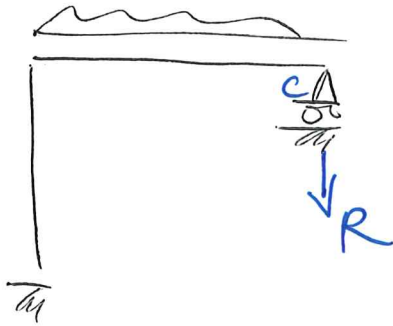
$$= \frac{1}{EI} \left(\int \square_{45} \cdot \square_3 dx + \int \triangle_3 \cdot \triangle_{45} dx \right)$$

$$= \frac{1}{EI} \left(3 \times 45 \times 3 + \frac{1}{4} \times 3 \times 45 \times 3 \right)$$

$$= \frac{2025}{4EI} (\downarrow)$$

Example 6

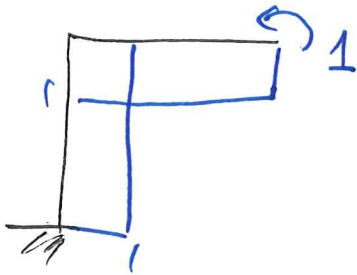
7/7



$$R = \frac{-\sum \int \frac{mM}{EI} dx}{\sum \int \frac{m^2}{EI} dx}$$

$$\begin{aligned} \sum \int \frac{m^2}{EI} dx &= \frac{1}{EI} \left(\int_0^3 \left[\begin{array}{c} \square \\ 3 \end{array} \right] \cdot \left[\begin{array}{c} \square \\ 3 \end{array} \right] dx + \int_0^3 \left[\begin{array}{c} \triangle \\ 3 \end{array} \right] \cdot \left[\begin{array}{c} \triangle \\ 3 \end{array} \right] dx \right) \\ &= \frac{1}{EI} \left(3 \times 3 \times 3 + \frac{1}{3} \times 3 \times 3 \times 3 \right) = \frac{36}{EI} \end{aligned}$$

$$R = - \frac{\frac{2025}{4EI}}{\frac{36}{EI}} = - \frac{225}{16} \approx -14.06 \quad (\uparrow)$$



$$\begin{aligned} \theta_c &= \frac{1}{EI} \left[\left(\int_0^3 \left[\begin{array}{c} \square \\ 1 \end{array} \right] \cdot \left[\begin{array}{c} \square \\ 45 \end{array} \right] dx + \int_0^3 \left[\begin{array}{c} \square \\ 1 \end{array} \right] \cdot \left[\begin{array}{c} \triangle \\ 45 \end{array} \right] dx \right) \right. \\ &\quad \left. + R \cdot \int_0^3 \left[\begin{array}{c} \square \\ 1 \end{array} \right] \cdot \left[\begin{array}{c} \square \\ 3 \end{array} \right] + \left[\begin{array}{c} \square \\ 1 \end{array} \right] \cdot \left[\begin{array}{c} \triangle \\ 3 \end{array} \right] dx \right] \\ &= \frac{1}{EI} \left(-45 \times 3 - \frac{45}{3} \times 3 + R \left(-3 \times 3 - \frac{3}{2} \times 3 \right) \right) \\ &= \frac{9.84}{EI} \quad (\curvearrowright) \end{aligned}$$